

Lem involucion disco unidad rho

Lema 1. Sea $f = \tau_a \circ \rho_\gamma \in \text{Aut}(\mathbb{D})$, entonces $f = \rho_\gamma \circ \tau_{\bar{\gamma}a}$.

Demostración: Como $\gamma \in \mathbb{T}$, tenemos que $\gamma \cdot \bar{\gamma} = |\gamma|^2 = 1$. Por tanto,

$$\forall z \in \mathbb{D} : f(z) = \tau_a(\rho_\gamma(z)) = \frac{a - \gamma z}{1 - \bar{a}\gamma z} \cdot \frac{\bar{\gamma}}{\gamma} = \frac{1}{\bar{\gamma}} \cdot \frac{\bar{\gamma}a - z}{1 - \bar{a}\gamma z} = \gamma \cdot \frac{\bar{\gamma}a - z}{1 - (\bar{\gamma}a)z} = \rho_\gamma(\tau_{\bar{\gamma}a}(z)). \quad \blacksquare$$